

האוניברסיטה העברית בירושלים
THE HEBREW UNIVERSITY of JERUSALEM
الجامعة العبرية في القدس

Introduction to Algorithms

1 Peaks Finding

Example 1. Consider an n-length array A such that $A_1, A_2, \ldots, A_n \in \mathbb{R}$. We will say that A_j is a **peak** (local maximum) if it is greater than its neighbors. Namely, $A_i \geq A_{i\pm1}$ if $i \pm 1 \in [n]$. Whenever $i \pm 1$ is not in the range $[n]$, we will define the inequality $A_i \geq A_{i\pm1}$ to hold trivially. For example, for $n = 1$, $A_1 = A_n$ is always a peak.

Write an algorithm that, given A, returns the position of an arbitrary peak.

Example 2. Warming up. How many peaks do the following arrays contain?

1. $A[i] = 1 \; \forall i \in [n]$
2. $A[i] = \begin{cases} i & i < n/2 \\ n/2 - i & \text{else} \end{cases}$
3. $A[i] = i \; \forall i \in [n]$

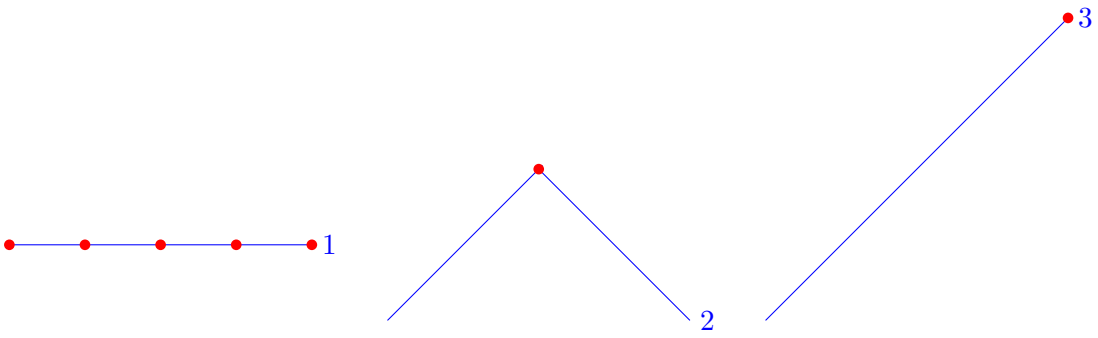


Figure 1: Illustration of the arrays on the smooth line, the peaks are marked in red.

2 Naive solution

To better understand the problem, let's first examine a simple solution before proposing a more intriguing one. Consider the algorithm examining each of the items A_i one by one.

Algorithm 1: naive peak-find alg.

Result: returns a peak of $A_1 ... A_n \in \mathbb{R}^n$

```
1 for i ∈ [n] do
2     if A_i is a peak then
3         return i
4     end
5 end
```

Correctness. We will say that an algorithm is correct with respect to a given task if it computes the task for any input. Let's prove that the above algorithm is doing the job.

Proof. Denote by j the position of the first peak in A, we will prove that Line 5 returns A_j.

Denote by i the current iteration step[1], and separate upon the following cases:

1. If $i < j$, then by the minimality of j (A_j was defined to be the first peak), it has to hold that A_i is not a peak. Therefore, the condition on line (2) is not satisfied $\Rightarrow$ the algorithm skips to line (4) and from there continues to the next iteration.

2. If $i = j$, then by definition of A_j, the condition on line (2) is satisfied, so the algorithm enters line (3) and returns A_j.

3. Else, consider the case when $i > j$. In that case, the algorithm will not reach the ith iteration. That is because the algorithm is defined to scan the elements in 'continuous' order, meaning that if the algorithm indeed gets into the i iteration, it implies that it passed through any $i' < i$ iteration and didn't return, specifically at the j iteration, which contradicts A_j being a peak.

□

Note that in the third bullet, in the case $i > j$, we proved a general claim about the behavior of simple 'for loop' programs. From now on, whenever we analyze a similar 'loop' that scans elements iteratively, we will not consider cases beyond the returning point and will be satisfied with proving analogous cases to 1 and 2. Also, in the exam/exercises, unless it is the main issue of the question, you are not expected to treat that case. (Spend your time wisely on what really matters.)

Running Time. Question, How would you compare the performance of two different algorithms? What will be the running time of the naive peak-find algorithm? At the lecture, you will see a well-defined way to treat such questions, but for the sake of getting the general picture, let's assume that we pay for any comparison quanta of processing time and, overall, checking if an item in a given position is a peak, cost at most $c \in \mathbb{N}$ time, a constant independent on n.

[1] In our course, in terms of terminology, usually 'step' = 'iteration' = 'iteration number' = 'turn' = 'index'.

Question, In the worst case scenario, how many local checks does peak-finding do? For the third example in Example 2, the naive algorithm will have to check each item, so the running time adds up to at most $c \cdot n$.

3 Naive alg. recursive version

Now, we will show a recursive version of the navie peak-find algorithm for demonstrating how correctness can be proved by induction.

Algorithm 2: naive recursive peak-find alg.

Result: returns a peak of $A_1 ... A_n \in \mathbb{R}^n$

```
if A_1 ≥ A_2 or n = 1 then
    return 1
end
return 1 + peak-find(A_2, ..A_n)
```

Claim 1. *Let $A = A_1, \ldots, A_n$ be an array, and*

$$A' = A'_1, A'_2, \ldots, A'_{n-1} = A_2, A_3, \ldots, A_n$$

be the $n-1$ length array obtained by taking all of A's items except the first. If $A_1 < A_2$,[2] then any peak of A' is also a peak of A.

Proof. Let A'_j be a peak of A'. Split into cases upon on the value of j.

1. If $n-1 > j > 1$, then $A'_j \geq A'_{j\pm1}$, but for any $j \in [2, n-2]$ we have $A'_j = A_{j+1}$ and therefore $A_{j+1} \geq A_{j+1\pm1} \Rightarrow A_{j+1}$ is a peak in A.
2. If $j' = 1$, then $A'_1 \geq A'_2 \Rightarrow A_2 \geq A_3$ and by combining the assumption that $A_1 < A_2$ we have that $A_2 \geq A_1, A_3$. So $A_2 = A'_1$ is also a peak.
3. The last case $j = n-1$ is left as an exercise.

□

One can prove a much more general claim by following almost the same argument presented above.

Claim 2. *Let $A = A_1, \ldots, A_n$ be an array, and $A' = A_{j+1}, A_{j+2}, \ldots, A_n$ be the $n-j$ length array. If $A_j \leq A_{j+1}$, then any peak of A' is also a peak of A.*

We are ready to prove the correctness of the recursive version by induction using Claim 1.

1. Base. Single element array. Trivial.
2. Assumption. Assume that for any m-length array, such that $m < n$ the algorithm returns a peak.

[2] weak inequality ($\leq$) is sufficient.

3. Step. Consider an array A of length n. If A_1 is a peak, then the algorithm answers affirmatively on the first check, returning 1 and we are done. If not, namely $A_1 < A_2$, then by using Claim 1 we have that any peak of $A' = A_2, A_3, \ldots, A_n$ is also a peak of A.

   The length of A' is $n - 1 < n$. Thus, by the induction assumption, the algorithm succeeds in returning on A' a peak which is also a peak of A.

4 An attempt for improved solution.

We saw that we can find an arbitrary peak at $c \cdot n$ time, which raises the question, can we do better? Do we really have to touch all the elements to find a local maxima? Next, we will see two attempts to catch a peak at logarithmic cost. The first attempt fails to achieve correctness, but analyzing exactly why will guide us on how to come up with both an efficient and correct algorithm.

Algorithm 3: fail attempt for more improved alg.

Result: returns a peak of $A_1 ... A_n \in \mathbb{R}^n$

```
1 i ← ⌈n/2⌉
2 if A_i is a peak then
3 |   return i
4 end
5 else
6 |   return i − 1+ find-peak(A_i, A_{i+1}..A_n)
7 end
```

Let's try to 'prove' it.

1. Base. Single element array. Trivial.

2. Assumption. Assume that for any m-length array, such that $m < n$ the algorithm returns a peak.

3. Step. If $A_{n/2}$ is a peak, we're done. What happens if it isn't? Is it still true that any peak of $A_i, A_{i+1}, \ldots, A_n$ is also a peak of A? Consider, for example, $A[i] = n - i$.

5 Improved solution

The reason we didn't succeed, and actually do not have a chance, to complete the induction step is that we didn't enforce any relation between the subarray A' and its ancestor A. We would like to have a claim similar to Claim 1. That brings us to define Line 10 and use Claim 2 to prove its correctness. Let's prove correction.

Proof. By induction.

1. Base. Single element array. Trivial.

2. Assumption. Assume that for any m-length array, such that $m < n$ the algorithm returns a peak.

Algorithm 4: improved algorithm

Result: returns a peak of $A_1 ... A_n \in \mathbb{R}^n$

```
i ← ⌈n/2⌉
if A_i is a peak in A then
    return i
end
else if i = 1 or A_{i-1} ≤ A_i then
    return i+ find-peak(A_{i+1}..A_n)
end
else
    return find-peak(A_1, A_2, A_3..A_{i-1})
end
```

3. Step. Consider an array A of length n. If $A_{\lceil n/2 \rceil}$ is a peak, then the algorithm answers affirmatively on the first check, returning $\lceil n/2 \rceil$ and we are done.

   If not, then either $A_{\lceil n/2 \rceil} < A_{\lceil n/2 \rceil - 1}$ or $A_{\lceil n/2 \rceil} < A_{\lceil n/2 \rceil + 1}$. We have already handled the first case, that is, using Claim 2 we have that any peak of $A' = A_{\lceil n/2 \rceil + 1}, A_{\lceil n/2 \rceil + 2}, \ldots, A_n$ is also a peak of A. The length of A' is $n - \lceil n/2 \rceil - 1 \leq \lfloor n/2 \rfloor < n$. So by the induction assumption, in the case where $A_{\lceil n/2 \rceil} < A_{\lceil n/2 \rceil - 1}$ the algorithm returns a peak.

   In the other case, we have $A_{\lceil n/2 \rceil} < A_{\lceil n/2 \rceil + 1}$ (otherwise $A_{\lceil n/2 \rceil}$ would be a peak).

   Finishing the proof is left as an exercise.

□

What's the running time? Denote by $T(n)$ an upper bound on the running time. We claim that $T(n) \leq c_1 \log(n) + c_2$.

Proof. By induction.

1. Base. For the base case, $n \leq 3$ we get that $c_1 \log(1) + c_2 = c_2$ on the other hand only a single check made by the algorithm, so indeed the base case holds (Choosing $c_2 \geq$ the cost of a single check).

2. Induction Assumption. Assume that for any $m < n$, the algorithm runs in at most $c_1 \log(m) + c_2$ time.

3. Step. Notice that in the worst case, $\lceil n/2 \rceil$ is not a peak, and the algorithm calls itself recursively immediately after paying c_2 in the first check.

   The recursive call is over an array length of $\lceil n/2 \rceil - 1$ [3]. Hence:

$$\begin{aligned} T(n) &\leq c_2 + T(n/2) \leq c_2 + c_1 \log(\lfloor n/2 \rfloor) + c_2 \\ &= c_1 - c_1 + c_2 + c_1 \log(\lfloor n/2 \rfloor) + c_2 \\ &= c_1 \log(2) + c \log(\lfloor n/2 \rfloor) + 2c_2 - c_1 \\ &= c_1 \log(2(\lfloor n/2 \rfloor)) + 2c_2 - c_1 \end{aligned}$$

[3] Notice $\lceil n/2 \rceil - 1 \leq \lfloor n/2 \rfloor$.

So, choosing $c_1 > c_2$ gives $2c_2 - c_1 < c_2$ and therefore:

$$T(n) \leq c_1 \log(n) + c_2$$

□